

CAMERA.APP

CS 180 - PROJECT 0

BECOMING FRIENDS WITH YOUR CAMERA

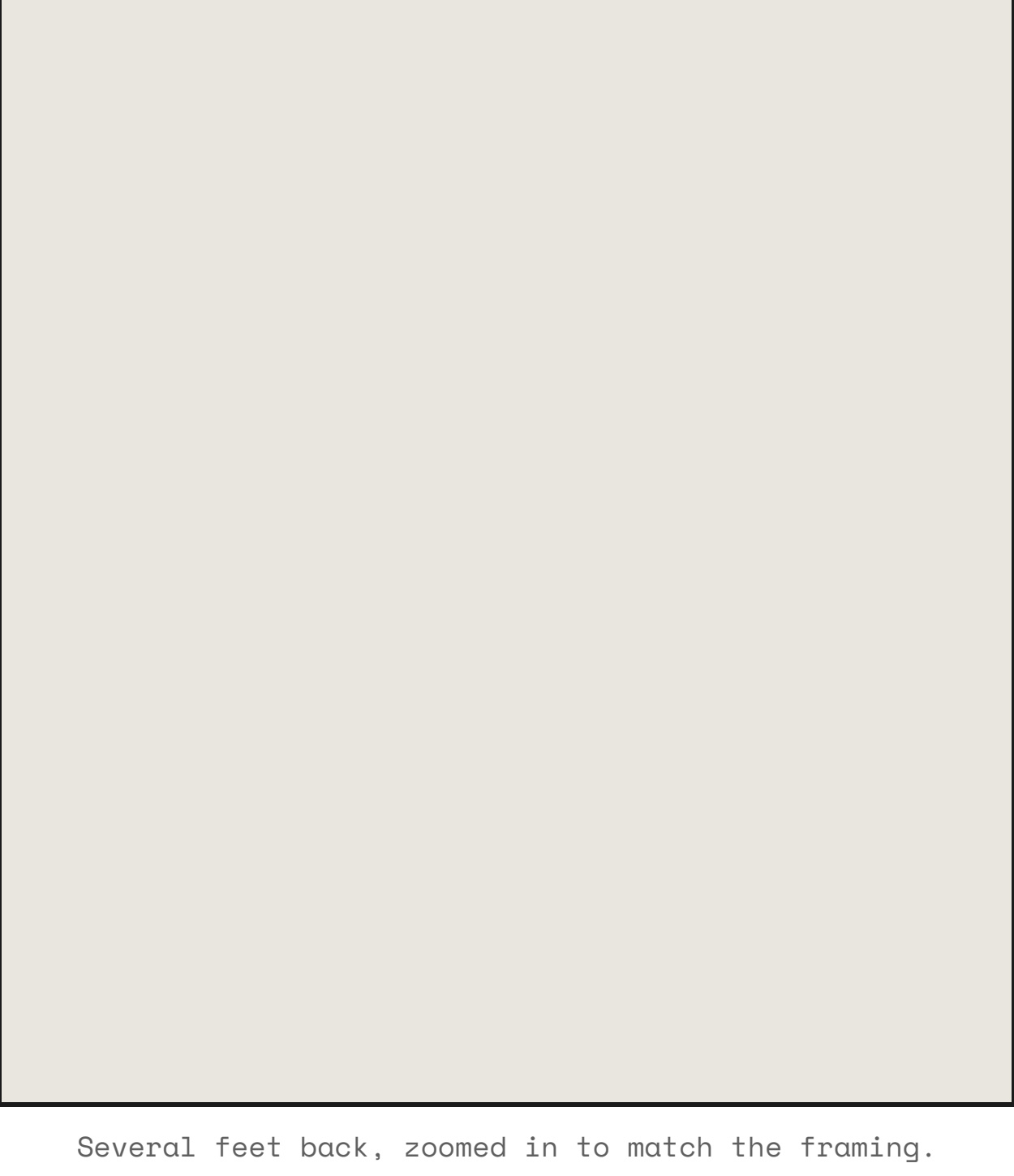
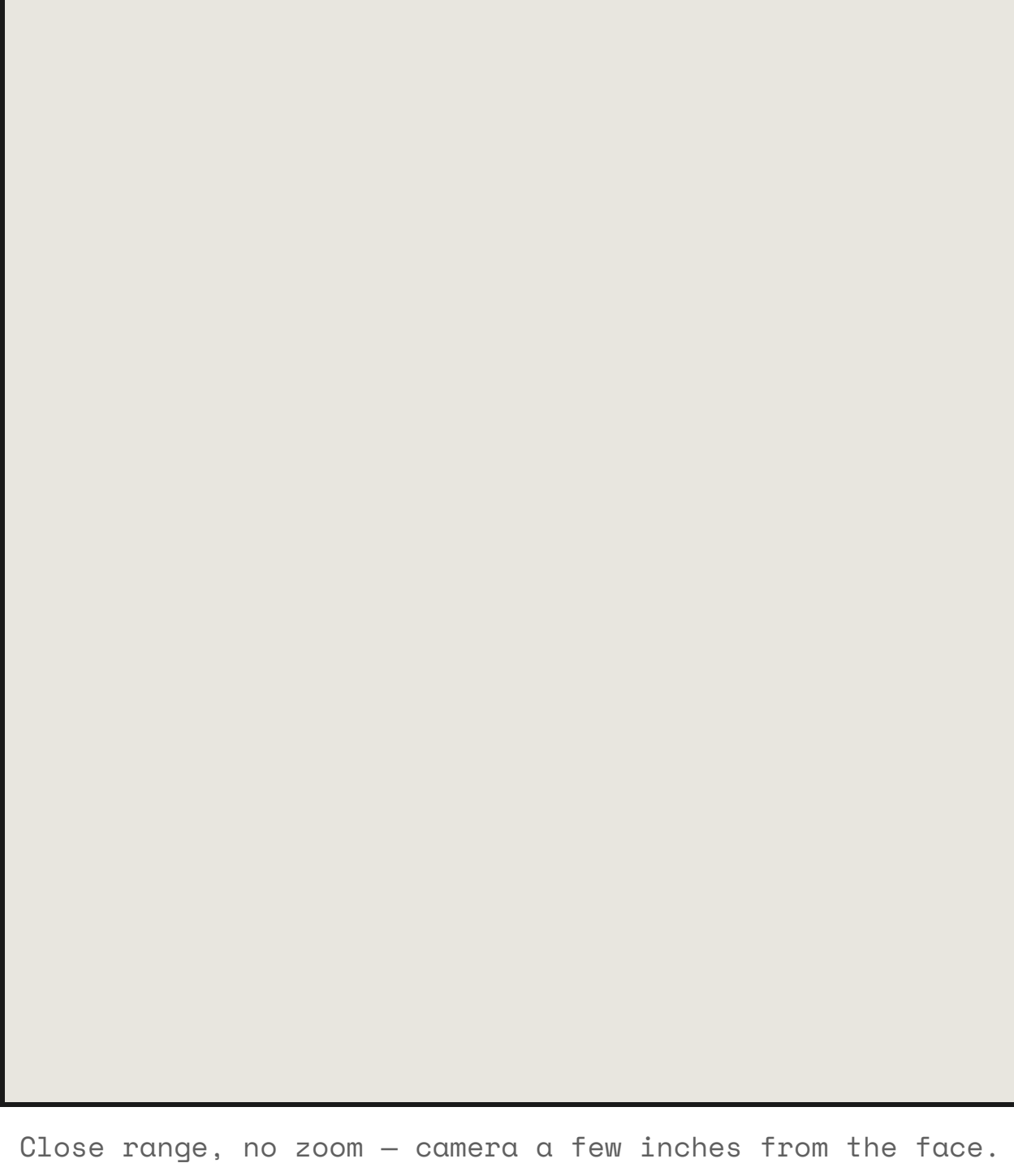
by Sohum Phadke / Fall 2026

READ ME

In this project, I explored the relationship between perspective, focal length/zoom, and the center of projection. Every camera makes two decisions at once: how far it stands from the world, and how much it magnifies what it sees. Change them together and a photo can stay the same size but tell a completely different story about depth. The three experiments below isolate that relationship – first on a face, then on a building, then in one continuous move known as the dolly zoom.

PART 1 – THE SELFIE: WRONG WAY VS. RIGHT WAY

CLOSE-UP.JPG VS. STEPPED-BACK-ZOOMED.JPG



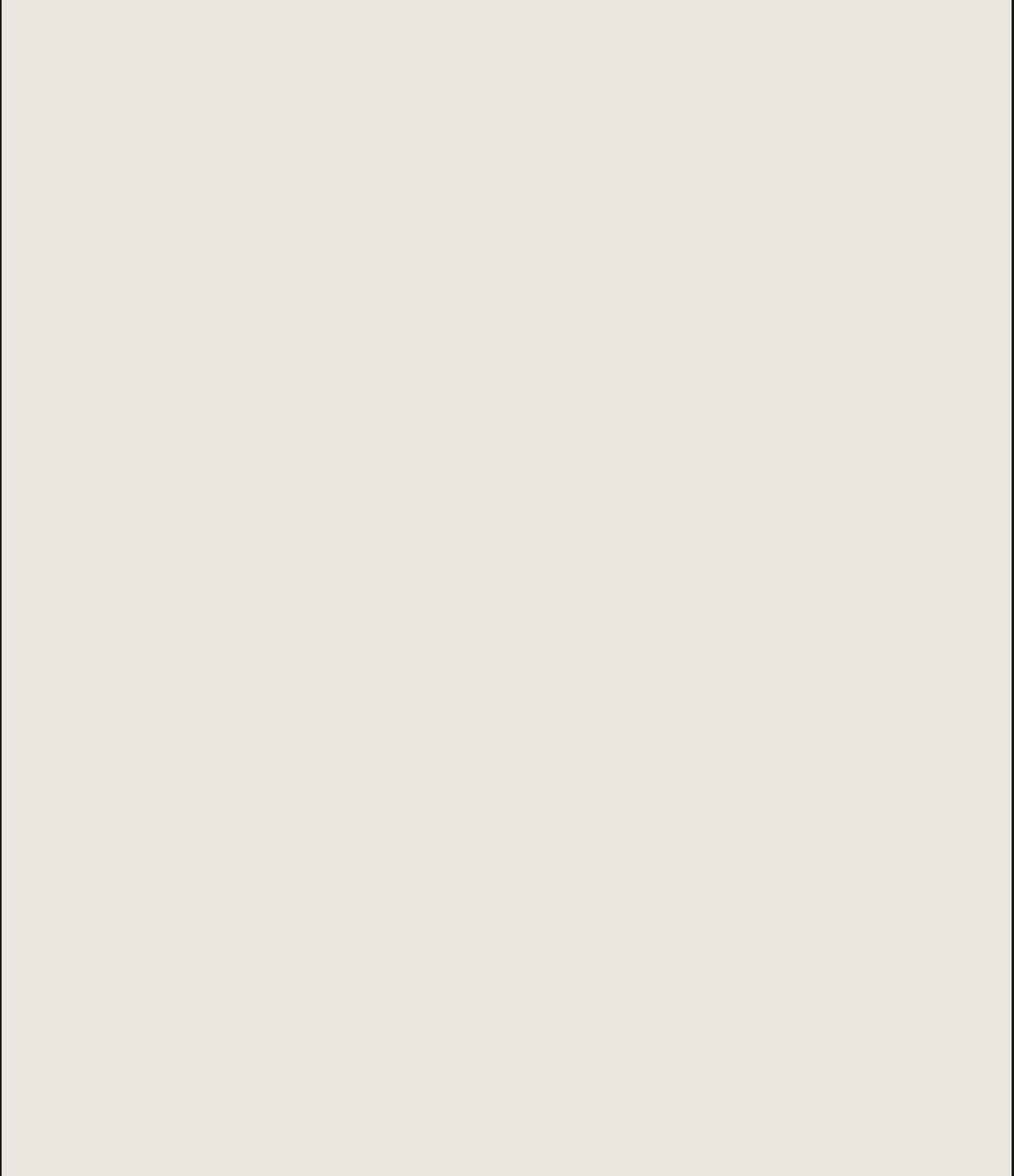
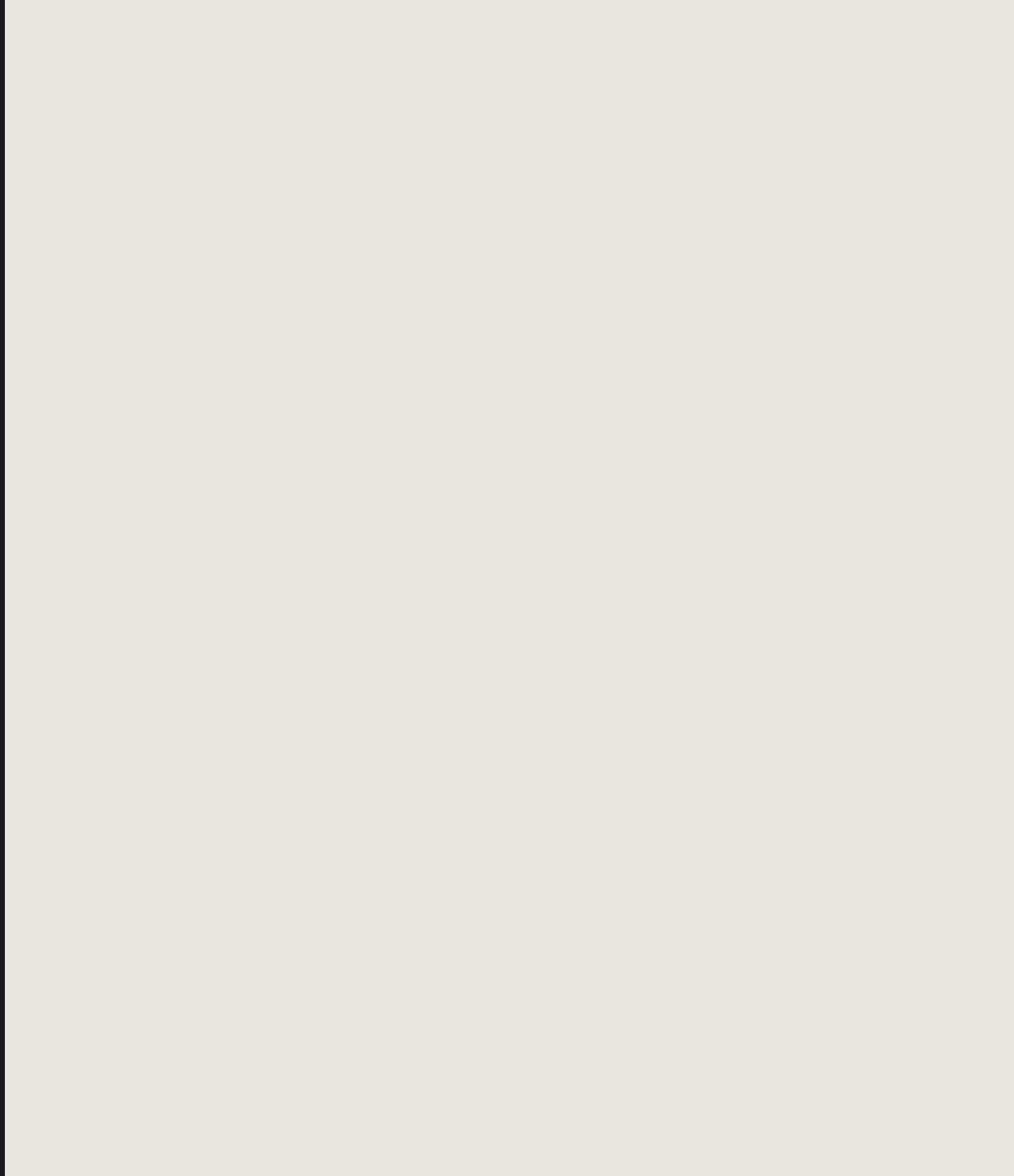
Close range, no zoom – camera a few inches from the face.

Several feet back, zoomed in to match the framing.

Moving the camera back a few feet and zooming in, while keeping my face the same size in frame, noticeably changed my proportions. Up close, my nose is much closer to the lens than my ears, so that small depth difference gets exaggerated: my nose looks bigger, my ears shrink and nearly vanish from frame, and my whole face looks rounder. Stepping back evens out the nose-to-lens and ear-to-lens distances, so everything looks more proportional, closer to how I actually look in person. It's the same reason portrait photographers shoot from farther away with a longer lens instead of getting close with a wide one.

PART 2 – ARCHITECTURAL PERSPECTIVE COMPRESSION

FAR-ZOOMED-IN.JPG VS. CLOSE-NO-ZOOM.JPG



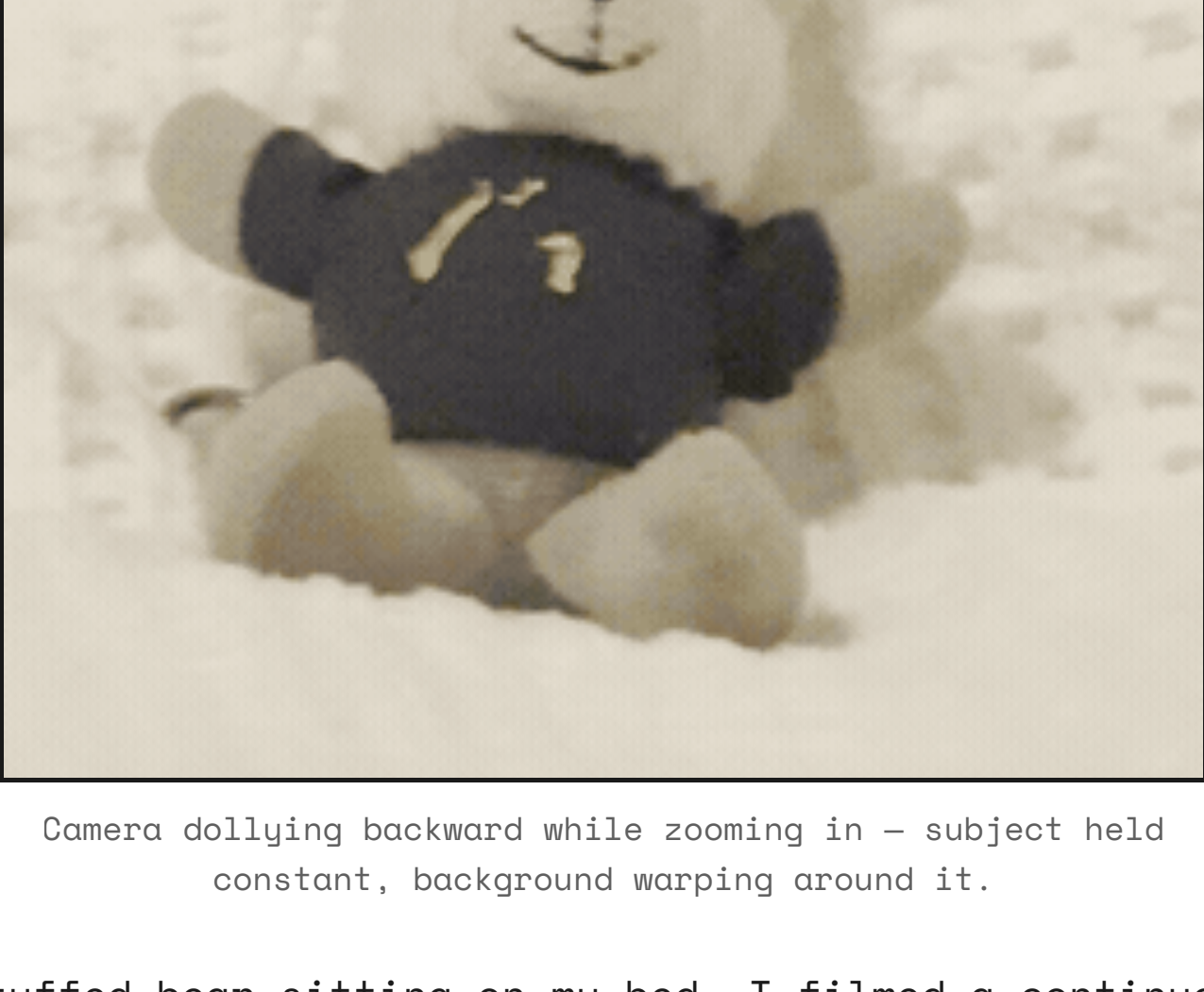
Far away, zoomed in – the compressed look.

Walked in close, no zoom – matched framing.

Zoomed in from far away, the building looks flattened, almost like a 2D grid, with the window lines staying close to parallel. Up close without zoom, the perspective converges hard: lines pull toward a vanishing point and the facade looks like it's dramatically receding. That's because the near and far parts of the building sit at very different distances from me up close (a large proportional difference), but from far away that same depth range becomes a tiny fraction of the total distance, so everything ends up looking roughly the same distance away and the facade compresses flat.

PART 3 – THE DOLLY ZOOM

DOLLY-ZOOM.GIF



Camera dollyng backward while zooming in – subject held constant, background warping around it.

My subject was a small stuffed bear sitting on my bed. I filmed a continuous video while walking backward and zooming in at the same time, keeping the bear the same size in frame throughout. The bear itself stayed fixed in size and position, but the headboard and pillow behind it visibly stretched and shifted as the background compression changed. That mismatch, subject constant while the background warps, is what makes a dolly zoom read as unsettling: your brain expects depth cues to move together, and here they're deliberately decoupled.

THREE STILLs FROM THE SEQUENCE



Start – wider shot, less zoom.

Midway – background starting to shift.

End – background fully compressed in.

CS 180 – Project 0: Becoming Friends with Your Camera

Made with a real camera, a bit of walking back and forth, and HTML.